Credit Application Governance Analysis.

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Unchecked Automation in Lending

How data quality failures and algorithmic bias expose NovaCred to regulatory risk.

CONTEXT

NovaCred is a fintech startup that uses machine learning to **automatically approve or reject credit applications**.

Following a **regulatory inquiry into potential discrimination**, our team was engaged as an **independent data governance task force**.

OUR MANDATE

1. Audit 502 credit applications for data quality failures
2. Detect algorithmic bias in lending outcomes
3. Assess GDPR & EU AI Act compliance gaps
4. Propose concrete governance controls

KEY FINDINGS

- Dataset fails all 6 quality dimensions
  → decisions on corrupt data
- Gender DI ratio = 0.77 (threshold: 0.80)
  → potential unlawful discrimination
- SSNs, emails & names stored in plaintext
  → GDPR Art. 5 violation
- 81.7% of rejections carry only 'algorithm_risk_score'
  → black-box, non-auditable under EU AI Act

Inconsistencies & Remediation

Gender encoding had 6 distinct values; date formats were split across 3 different standards.

Gender BEFORE Standardization

Gender	Count
Male	195
Female	193
F	58
M	53
""	2
NaN	1

→ *Normalize*

AFTER Standardization

Gender	Count
Male	248
Female	251
""	2
NaN	1

Date Format Issues

Format	Count	%
YYYY-MM-DD	340	67.7%
MM/DD/YYYY	101	20.1%
YYYY/MM/DD	56	11.2%
<MISSING>	5	1.0%

pd.json_normalize() to extract nested attributes, followed by a regex-based diagnostic to identify inconsistent date strings before standardizing them into a unified ISO 8601 format.

→ Without standardization, **basic demographic analysis (age, gender fairness) would produce unreliable results.**

4 Categories of Quality Issues

All issues were identified, quantified, and remediated in notebook 01-data-quality.ipynb

Missing / Incomplete

497/502

Around 99% of records

Column with the most missing values:
notes and annual_salary

Inconsistent Data Types

157/502

31.3% of records

Rows failing date parsing before fix → 0 after normalization

Duplicate Rows

2/502

1.42% of records

After flattening nested JSON columns

Invalid Values

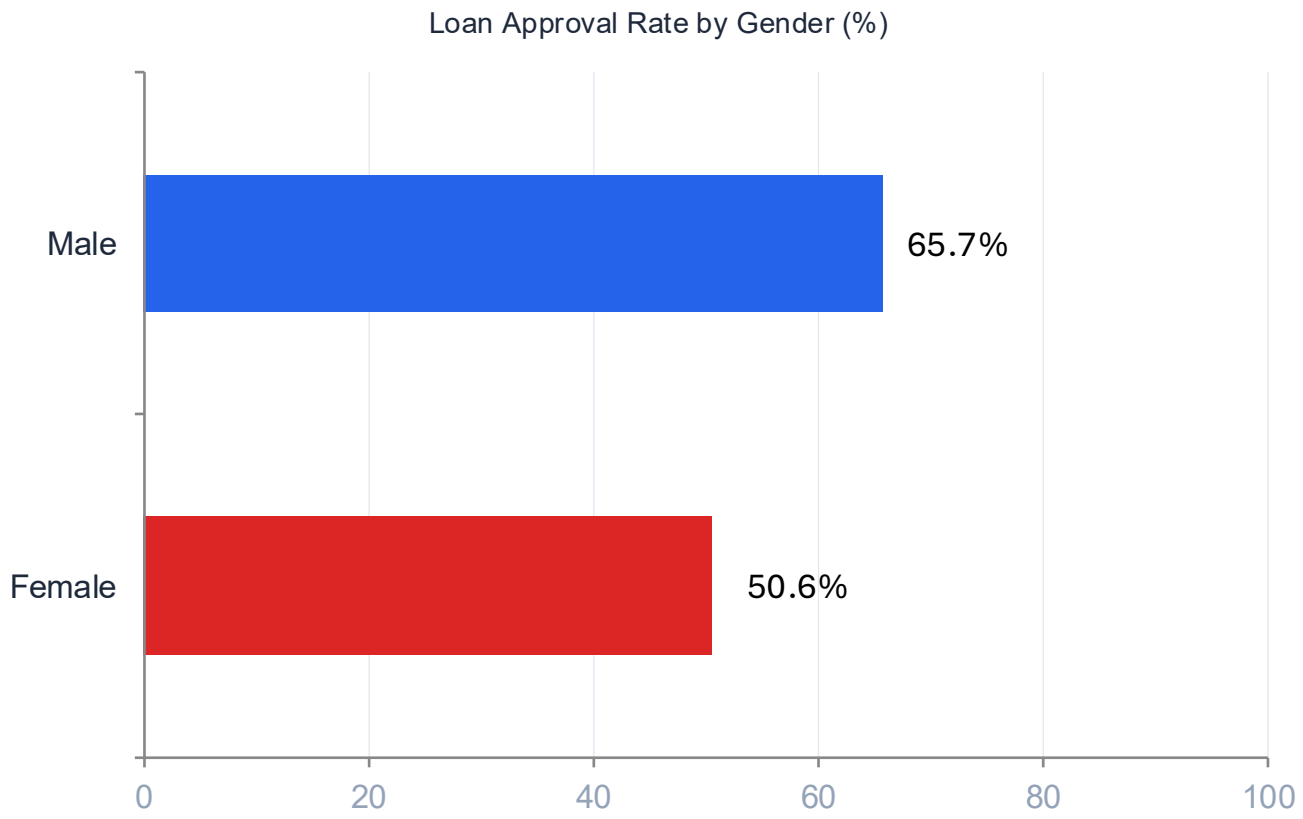
8/502

1.6% of records

Missing gender = 3 (0.6%)
Credit history = 2 (0.4%)
Savings balance = 1 (0.2%)
Debt-to-income = 1 (0.2%)
Annual income = 1 (0.2%)

Does NovaCred Discriminate?

Four-Fifths Rule (80% Rule): DI below 0.80 indicates potential unlawful disparate impact.



Disparate Impact Ratio

0.770

⚠ BELOW 0.80 THRESHOLD — FAIL

KEY FINDINGS

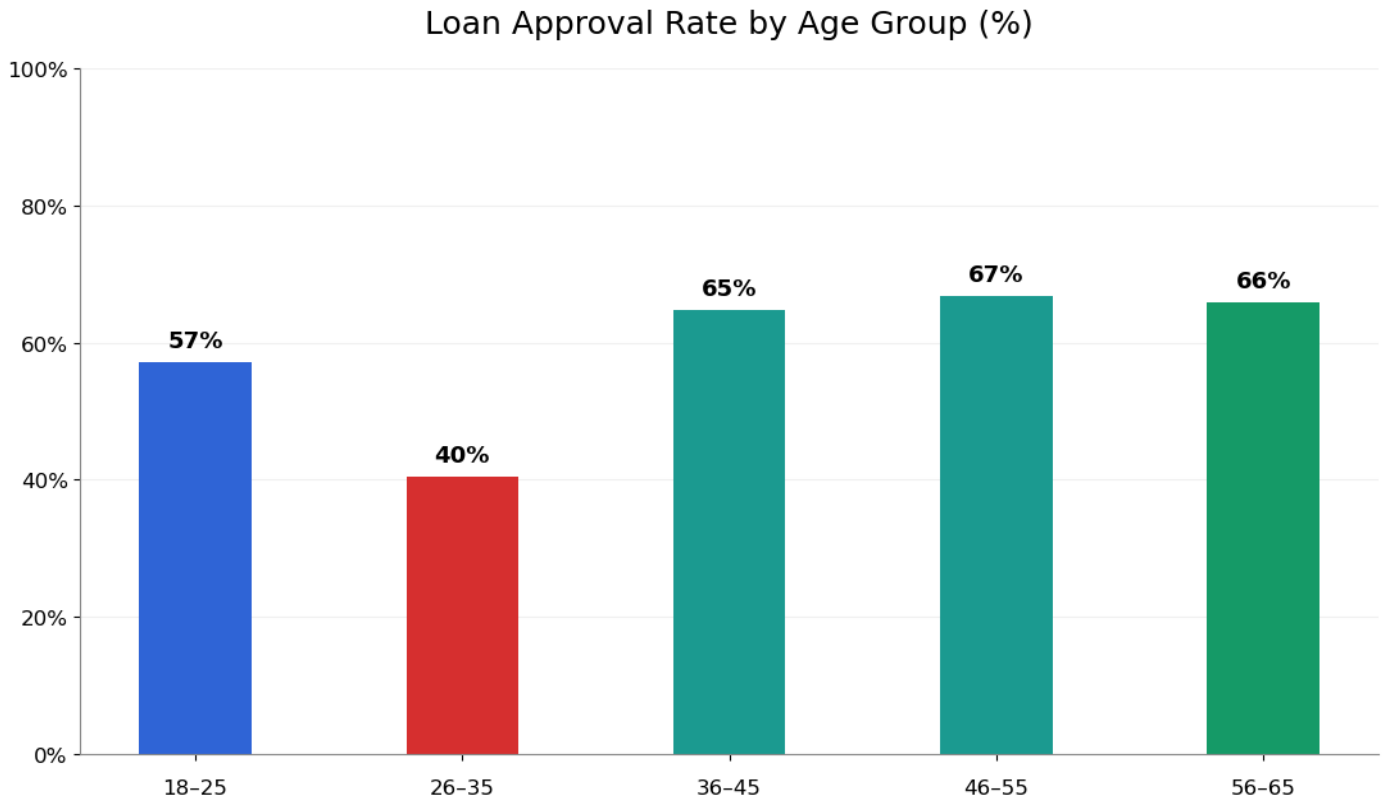
Female applicants approved at **50.6%** vs **65.7%** for males

DI = 0.506 ÷ 0.657 = **0.770**, therefore below the 0.80 threshold

Implication: Potential gender-based discrimination that requires investigation under the EU AI Act.

Hidden Bias Beyond Gender

Age patterns and proxy variables amplify discrimination even when gender is excluded from the model.



ZIP Code Proxy

zip_code correlates with gender at r = −0.806 (point-biserial). Geographic filtering reproduces gender discrimination even after gender is removed from the model.

Age-Group Disparities

Applicants aged 26–35 have the lowest approval rate (~40%). Applicants (46–55) have the highest (~67%)

Intersection Effect

Women aged 26–35: ~33% approval. Men aged 56–65: ~78% approval. This highlights intersectional disparities when age and gender combine.

Personal Data Risks in the Dataset

Multiple GDPR violations identified: no pseudonymization, no consent, no retention policy.

PII INVENTORY

Direct identifiers: `full_name, email, ssn, ip_address`

Indirect identifiers: `date_of_birth, zip_code, gender`

PSEUDONYMIZATION DEMO

Step	Value	Risk
Original	`596-64-4340`	**Full exposure**
SHA-256	`2caf30528c21a10e13...`	**Bruteforceable**
Salted hash	`8d15601299eaf099f4...`	**Secure**

GDPR COMPLIANCE GAPS

Art. 5(1)(c) **Data Minimization:**

ip_address undocumented; 15 spending categories reveal sensitive lifestyle data.

Art. 6 **Lawful Basis:**

0/502 records have consent, processing purpose, or data source documented.

Art. 17 **Right to Erasure:**

No retention_until or created_at fields; no deletion mechanism exists.

EU AI ACT CLASSIFICATION

Credit scoring = HIGH-RISK AI (Annex III 5b)

Subject to strict requirements: data quality (Art. 10), transparency (Art. 13), human oversight (Art. 14), and record-keeping (Art. 12).

What NovaCred Must Do

Four priority actions to achieve regulatory compliance and prevent future violations.

1 Pseudonymize All Direct Identifiers

ssn, full_name, email, ip_address stored in plain text. A single breach exposes all 502 applicants irreversibly. Apply salted SHA-256 hashing immediately.

Addresses: GDPR Art. 5(1)(f), Art. 25

2 Implement Ongoing Bias Monitoring

DI ratio of 0.77 shows gender discrimination. Establish monthly DI monitoring by gender and age. Any metric below 0.80 must trigger mandatory human review.

Addresses: EU AI Act Art. 9, Art. 10

3 Replace Black-Box Rejection Labels

81.7% of rejections carry only 'algorithm_risk_score' — non-interpretable. Applicants cannot understand or contest decisions, violating GDPR Art. 22 and AI Act Art. 13.

Addresses: GDPR Art. 22; EU AI Act Art. 13

4 Enforce Validation Schemas at Ingestion

6 gender encodings, 3 date formats, strings in numeric fields all entered unchecked. Every downstream issue — bias, missing data, invalid values — originates here.

Addresses: EU AI Act Art. 10

Six Metrics. One Verdict

NovaCred's automated lending system fails every dimension of responsible AI governance.

6/6
Quality dimensions failed

0.770
Disparate Impact (threshold: 0.80)

−0.806
zip_code ↔ gender proxy correlation

81.7%
Rejections with no explanation

7 PII
Fields in plaintext (incl. SSNs)

0/502
Records with consent or retention

NovaCred is operating a high-risk AI system (EU AI Act, Annex III §5b) that violates GDPR Art. 5 on every principle, discriminates against female applicants, and cannot explain its own decisions.

Thank you!

Any Questions?

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